The Selling closure

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Abstract

A lattice with a right angle has several Selling-reduced cells; Kurlin (2022) showed that there are at most sixteen. This paper treats the set of all of them, with the integer changes of basis between them, as one object, the Selling closure. The closure contains every lattice symmetry and every operator between alternative reduced cells. Symmetries relate only cells of identical shape, and in the primitive lattice types with two or three right angles most of the point group is not visible in any single reduced cell: none of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell. The members and operators of the closure are fixed by which conorms vanish, which operators are symmetries by which are equal, and the 24 Delone sorts are the closed patterns of equalities under that correspondence. For measured cells, near-congruence within a conorm band τ is defined and shown to be decided by a widened closure, Selling reduction with its boundary moved to $-\tau$. The widened closure is unique and is the union of the exact closures of the lattices within the band, the band being measured in each member's own conorms. When the shortest lattice vector satisfies

$\lambda_1 > 32\tau$ it is computed completely by setting every conorm of at most 4τ to zero, enumerating the exact closure there, and filtering with the measured metric; walks from the measured cell miss members. From a single reduced cell of a new cell the reindexing operator onto a reference is then found, with a stated guarantee.

1. Introduction

Two questions recur in indexing and data processing. What are the symmetries of a lattice? And are two cells the same lattice, and if so by which integer matrix? Both are usually answered by searching: for symmetry, a scan for near-twofolds within an angular tolerance (Le Page, 1982); for reindexing, a list of small integer matrices, or the operators of the point group just found. The symmetry search is complete within its index bound; the reindexing search is complete only if the list happens to contain the operator. Both are complicated by the same fact: a lattice with a right angle has more than one reduced cell, and the alternatives are related by integer matrices that are not symmetries.

Selling reduction makes the multiplicity explicit. A lattice on a right angle has several obtuse superbases, as Delaunay (1933) noted and Patterson & Love (1957) made practical; Kurlin (2022) proved that there are never more than sixteen and listed them by zero pattern. We take the whole set, with the integer matrices between its members, as the object of study and call it the *Selling closure*; the term is introduced here. Section 2 defines it and works through one monoclinic cell; Section 3 states the computations; Section 4 gives what the closure contains, how it is computed for a measured cell, and what that gives for reindexing. Proofs are in Appendix A, the conversion of a measurement error into a band in Appendix B, catalogues that the main text only refers to in Appendix C, a description of the operators with no lattice in Appendix D, and every computed number in the text carries a mark [znn] naming

the check that reproduces it (Appendix E).

2. The Selling closure

2.1. Superbases, conorms and the metric tensor

Selling reduction works with four vectors rather than three: the cell vectors a, b, c and $d = -a - b - c$. The four form a *superbase*; any three of them are a basis of the lattice. The six negated pair products

$$p_{da} = -d \cdot a, \quad p_{db} = -d \cdot b, \quad p_{dc} = -d \cdot c, \quad p_{ab} = -a \cdot b, \quad p_{ac} = -a \cdot c, \quad p_{bc} = -b \cdot c$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992); the Selling parameters of the S^6 formalism (Andrews *et al.*, 2019a,b) are the same numbers with the opposite sign. Conorm vectors are written in this slot order, da, db, dc, ab, ac, bc, throughout, except that the sort patterns of Table 3 use the S^6 order bc, ac, ab, ad, bd, cd so that they can be compared with Andrews & Bernstein (2023b) and Andrews *et al.* (2023). A superbase is *obtuse*, which is to say Selling-reduced, when all six conorms are non-negative. The metric tensor in the basis (a, b, c) is linear in the conorms,

$$\mathbf{G} = \begin{pmatrix} p_{ab} + p_{ac} + p_{ad} & -p_{ab} & -p_{ac} \\ -p_{ab} & p_{ab} + p_{bc} + p_{bd} & -p_{bc} \\ -p_{ac} & -p_{bc} & p_{ac} + p_{bc} + p_{cd} \end{pmatrix}, \tag{1}$$

so the conorms are an alternative set of six cell parameters, and a change to the conorms is a change to the cell with nothing to reconstruct. A conorm that is exactly zero is a right angle between two of the four vectors; this paper calls a zero conorm a *wall*, after the boundary it forms in the space of cells, and the set of slots that vanish the *zero pattern* of the cell. Only six zero patterns are possible if the four vectors are to remain connected by non-zero conorms: no zero; one; two sharing a vector; two disjoint; three forming a triangle; three forming a path. The last two occur in the same lattices — a lattice with three right angles has reduced cells of both kinds — so there

are five kinds of closure, which correspond to the five types of Voronoi polyhedron (Voronoi, 1908; Conway & Sloane, 1992).

2.2. The Selling operators

Two kinds of integer operator act on a superbase, and every operator in this paper is a product of them.

Flips. A flip at the conorm p_{ab} replaces a by $-a$ and adds the old a to the two other vectors: $(a, b, c, d) \mapsto (-a,\ b,\ c + a,\ d + a)$. It changes the sign of p_{ab}, and the new conorms are $p'_{ac} = p_{ab} + p_{ad}$, $p'_{ad} = p_{ab} + p_{ac}$, with p_{bc}, p_{bd}, p_{cd} unchanged; so a flip at a conorm that is exactly zero merely exchanges two conorms, and all reduced cells of one lattice carry the same six conorm values. The vector negated is the first of the pair in the order d, a, b, c: a flip at da, db or dc negates d. Selling reduction applies flips at negative conorms until none remain; the sum of the conorms decreases at each step, so it stops, at an obtuse superbase. A flip at a conorm that is exactly zero carries one obtuse superbase to another obtuse superbase of the same lattice, which is why the reduced cell of a lattice with a right angle is not unique. With the basis (a, b, c) each of the six flips is a fixed 3×3 integer matrix of determinant -1 that is its own inverse (Table 1); a word of flips such as *ab.bc.db* is applied left to right, each flip read in the current labelling, and its matrix is the product $\mathbf{F}_{ab}\mathbf{F}_{bc}\mathbf{F}_{db}$ of the matrices of Table 1, columns being the new basis vectors in the old basis. The flips are the boundary transformations of Andrews *et al.* (2019a), written on the basis rather than on the six Selling parameters.

Table 1. *The six Selling flips as* 3×3 *integer matrices in the basis* (a, b, c), $d = -a - b - c$; *columns are the new basis vectors in the old basis. Each has determinant* -1 *and squares to the identity. The second column gives the flip as a setting change in the notation of crystallographic software; for the flips at da, db, dc the vector d is negated and the new* a, b, c *are listed.*

flip at	new basis	matrix
ab	$(-a,\ b,\ a+c)$	$\begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
ac	$(-a,\ a+b,\ c)$	$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
bc	$(a+b,\ -b,\ c)$	$\begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
da	$(a,\ -a-c,\ -a-b)$	$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$
db	$(-b-c,\ b,\ -a-b)$	$\begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$
dc	$(-b-c,\ -a-c,\ c)$	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$

Swaps. A superbase is an unordered set of four vectors up to a common sign, so one reduced cell can be written in $4! \times 2 = 48$ ways. The 48 reorderings and sign changes are the swaps. They change which three vectors are called a, b, c and in which order, and nothing else. A swap is written by listing which vectors of the superbase become the new d, a, b, c, with a leading sign for the common sign change: $-(a, c, b, d)$ makes the old a the new d, the old c the new a, keeps b, makes the old d the new c, and negates all four. In matrices a swap is a signed permutation of the basis, with the fourth vector d allowed to enter it; the 48 swap matrices are written $\mathbf{Q}$ below.

Products, and symmetries among them. A sequence of flips at zero conorms followed by a swap carries a reduced cell to a reduced cell of the same lattice, possibly one of different shape, and every integer matrix that carries a reduced cell to a reduced cell is such a product (Kurlin, 2022; Lemma A7). A lattice symmetry is an integer matrix g with $g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}$ (Himes & Mighell, 1987); it carries reduced cells to reduced cells,

so it is one of these products. Which products are symmetries depends on the conorm values, not on the operators: a product is a symmetry for exactly those lattices whose conorms it leaves unchanged. The twofold rotations are the symmetries that matter in practice: together with the inversion they generate the point group of any lattice, which is the basis of the Le Page (1982) search, of the tolerance-ranked subgroups of Sauter *et al.* (2004), and of the twinning operators of Lebedev *et al.* (2006).

2.3. The closure, and one cell used throughout

The *Selling closure* $\mathcal{C}(\mathbf{G})$ of a lattice is the set of all its obtuse superbases, each counted once up to swaps, together with the products of flips and swaps between them. That the obtuse superbase is unique except when a conorm vanishes is Delaunay's (1933) observation, set out by Patterson & Love (1957) and by Conway & Sloane (1992); Kurlin (2022) proved that the set has 1, 2, 3, 6 or 16 members according to the pattern of zero conorms and that flips at zero conorms reach all of them. The members, as integer combinations of the reduced cell vectors, and the operators between them are the same for every lattice with the same zero pattern (Kurlin, 2022, Lemmas 4.1–4.5, which list them); only the conorm values differ. Two members have the same *shape* when their conorm vectors agree up to a swap; in its own basis each member is a cell, and members of different shape are differently shaped cells. The operators can also be characterised with no lattice at all, as the integer matrices that map a face of the orthant of non-negative conorms — the region where all six conorms are ≥ 0, a face being where a chosen set of them is 0 — into that orthant (Appendix D).

Take $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with conorms

$$p_{da} = 639.5, \quad p_{db} = 1600.0, \quad p_{dc} = 2239.5, \quad p_{ab} = 0, \quad p_{ac} = 260.5, \quad p_{bc} = 0 \text{ Å}^2.$$

The two walls are the right angles α and γ at the twofold axis b, present in every primitive monoclinic lattice. No other conorm is small, so the lattice is not near a higher symmetry.[z46] The closure has six members (Table 2); each has two walls, each flip leads to a neighbour, and the six form a ring M_0–M_1–M_5–M_4–M_3–M_2–M_0. All six carry the same six conorm values in different positions, and they fall into three shapes: $\{M_0, M_4\}$ is 30, 40, 50 Å, $\beta = 100°$; $\{M_1, M_3\}$ is 30, 40, 53.7 Å, $\beta = 113.4°$, the alternative choice $a + c$; $\{M_2, M_5\}$ is the third cell choice, on c and $a + c$.[z46]

Table 2. *The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: six obtuse superbases, each up to the order and common sign of its vectors, the flips at zero conorms that connect them, and where the four symmetries of the point group $2/m$ carry M_0.*[z46]

member	four vectors	zero-conorm flips lead to	symmetries carry M_0 here
M_0	$-a-b-c,\ a,\ b,\ c$	M_1, M_2	1, $\bar{1}$ (swaps of M_0)
M_1	$-b-c,\ -a,\ b,\ a+c$	M_0, M_5	
M_2	$-a-c,\ a+b,\ -b,\ c$	M_0, M_3	
M_3	$a+c,\ b-c,\ -b,\ -a$	M_2, M_4	
M_4	$a-b+c,\ -c,\ b,\ -a$	M_3, M_5	2 along b, m normal to b
M_5	$a-b,\ c,\ b,\ -a-c$	M_1, M_4	

The point group is $2/m$ and each of its four operations carries M_0 to a member (last column of Table 2).[z46] The identity and the inversion are swaps of M_0, visible in it. The twofold sends $(a, b, c) \mapsto (-a, b, -c)$ and so carries $(-a - b - c, a, b, c)$ to $(a - b + c, -a, b, -c)$: not a reordering of M_0 but the member M_4, three flips away, and the mirror does the same. Read in the twofold's order M_4 has the metric tensor of (a, b, c); read in the order of Table 2 it has a different one, with 639.5 and 2239.5 exchanged. As a product, the twofold is the flips ab, bc, db — the path $M_0 \to M_1 \to M_5 \to M_4$, which ends at M_4 in the labelling $(c,\ -a + b - c,\ -b,\ a)$ — followed by the swap $-(a, c, b, d)$, which gives $(a - b + c, -a, b, -c)$; the mirror is the same flips with the swap (a, c, b, d).[z46] Half of the point group is therefore not visible in the reduced cell: it consists of operators between reduced cells. "Visible" means only that the operation reorders the reduced superbase; a search over lattice vectors, as Le Page's, finds the twofold either way, since it tests directions rather than

relabellings.

2.4. Delone sorts

Delaunay (1933) classified lattices into 24 sorts by the pattern of zeros and equalities among the reduced Selling parameters, each sort lying in one Bravais type and some Bravais types comprising several sorts (Delone *et al.*, 1975; Burzlaff & Zimmermann, 1985, 2006); the 24 patterns, with the symbols C1 to A3 used below, are those of Andrews & Bernstein (2023b) and Andrews *et al.* (2023). The sorts are the Selling counterpart of the 44 Niggli lattice characters (Niggli, 1928; de Wolff & Gruber, 1991; de Wolff, 2006): they split a Bravais type by the shape of the cell, not by its symmetry. The closure gives the sorts without trial values. Each operator of a zero pattern permutes and recombines the six conorms, and it is a symmetry of a lattice exactly when it leaves that lattice's conorms unchanged; a sort is then a set of equalities among the conorms that is forced by its own symmetries and forces no more (Proposition A9). Enumerating those closed sets of equalities from the operators alone gives 8, 5, 3, 5 and 3 sorts for the five kinds of closure in the order of Section 2.1, 24 in all, and the symmetry orders come out as Delone's holohedries.[z53] An earlier derivation by closed tie patterns on generic lattices gives the same 24.[z48] Since the closure's members and operators depend only on the zero pattern and its symmetries on the equalities, the sort is the natural unit for describing closures.

2.5. Measured cells

A measured cell is never exactly on a wall: a right angle is measured as 90.25° and the conorm is a small number of either sign. The band τ, in Å^2, is the tolerance within which a conorm counts as zero; Appendix B says how a declared error η on the metric tensor, its Frobenius norm in the reduced frame, is converted into τ; the examples use

$\tau = 3\eta$. The *widened closure* of a metric at band τ is

$$\mathcal{S}_\tau(\mathbf{G}) = \left\{ F \ : \ \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \right\}, \tag{2}$$

the superbases that Selling reduction would accept with its boundary moved from 0 to $-\tau$. The *depth* of a superbase is the size of its most negative conorm, so it belongs to the widened closure when its depth is at most τ, and its depth is the band at which it enters. Two cells are τ-*congruent* when some reduced superbase of one, carried by an integer operator $\mathbf{P}$ onto the other, has each of its six conorms agreeing to within τ under the two metrics; exact congruence, $\mathbf{P}^\mathsf{T}\mathbf{G}\mathbf{P} = \mathbf{G}'$, is the case $\tau = 0$. Finally, λ_1 denotes the squared length of the shortest lattice vector, and the *price* of an operator $\mathbf{P}$ between two metrics is the residual $\|\mathbf{P}^\mathsf{T}\mathbf{G}\mathbf{P} - \mathbf{G}'\|_F$ with $\mathbf{G}' \approx \mathbf{P}^\mathsf{T}\mathbf{G}\mathbf{P}$; with $\mathbf{G}' = \mathbf{G}$ it is the amount by which $\mathbf{P}$ fails to be a symmetry.

3. Methods

3.1. Closures and symmetries by sort

For each of the 24 sorts a generic lattice was built from its pattern with random values for the free parameters. Its closure was enumerated by flips at the exact zeros from the reduced superbase, members being kept once up to swaps, and its symmetries found by testing every member, in every order and sign of its four vectors, against the metric tensor. Symmetries carrying the reduced superbase to itself were counted separately from those carrying it elsewhere. Members were grouped into shapes — the same conorm vector up to swaps — and into orbits under the symmetries. The sorts themselves were derived from the operators as described in Section 2.4 and Appendix A.

3.2. Measured cells: idealise, enumerate, filter

The widened closure of equation (2) is computed as follows.

1. Selling-reduce the measured cell to a superbase F_0.

2. Set every conorm of F_0 that is at most 4τ to zero, all at once, and form the metric tensor $\mathbf{G}^*$ of the result by equation (1). This is the *idealised lattice* Λ^*; it has an exact zero pattern. If $\mathbf{G}^*$ is singular, or λ_1 of the measured cell — the smallest squared length among a, b, c, d and $a+b, a+c, b+c$ of F_0 — is not above 32τ, the hypothesis of Theorem A6 (Appendix A) fails and the procedure stops here (Section 4.2).

3. Enumerate the exact closure of Λ^* by flips at its zeros, keeping each member once up to swaps as integer combinations of F_0 — equivalently, keeping the operators $\mathbf{W}_1, \ldots, \mathbf{W}_m$ from F_0 to each member.

4. Evaluate each member with the *measured* metric tensor and keep those whose six conorms are all at least $-\tau$. Record each member's depth.

The operators of step 3 are integer matrices independent of the metric, so they are the same for Λ^* and for the measured cell; the measured metric enters only in step 4. Section 4.2 states the theorem that this returns exactly $\mathcal{S}_\tau(\mathbf{G})$.

3.3. Reindexing from one reduced cell

A reference R is stored as its widened closure in integer coordinates, the operators $\mathbf{W}_k$ of step 3 that survive step 4. A centred reference is first taken to a primitive cell, and its centring matrix is composed with the operator found. A new cell N is reduced once, to a basis B_N in its own coordinates. For each stored $\mathbf{W}_k$ and each of the 48 swaps $\mathbf{Q}$, the candidate operator is $\mathbf{P} = \mathbf{W}_k \mathbf{Q}\, B_N^{-1}$, so that $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}} \mathbf{G}_R \mathbf{P}$, and candidates are ranked by their price in R's frame. Candidates come in pairs $\pm\mathbf{P}$, and those of determinant -1 are improper; for a chiral crystal they are discarded. Membership of the true operator in the candidate set is what is guaranteed; the price

does not single it out within the coset of the symmetries and pseudo-symmetries, which is a matter for the intensities. The band is set from the declared error η: the worked examples use $\tau = 3\eta$, which covers every stored member in the frames that occur there but is not a worst-case guarantee, and the 180-cell test of Section 4.3 uses the coverage bound τ_{cov} of Appendix B; with a covariance for the metric tensor the band can be set slot by slot.

3.4. Verification

Closure memberships were checked against brute force: every basis with coordinates in $[-2, 2]$ relative to F_0 (135 408 bases of determinant ± 1), completed to a superbase by its fourth vector, tested against equation (2). Every operator between reduced cells has a labelling whose three basis vectors lie in this window (Appendix D), so the window contains every member.[z52] Test lattices were specified by their conorms. The measured worked cell is 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80°. It is the cell 30–40–50 Å, $\beta = 92°$ — the cell of Section 2.3 with β moved close to 90°, so that it sits on two walls and near a third — perturbed by 14.4 Å^2 in the Frobenius norm of its metric tensor. Its six members correspond to M_0 to M_5. It is treated at $\tau = 43.2$ Å^2, a band chosen large enough to show the filter at work, and at $\tau = 20$ Å^2, within the hypothesis of Theorem A6. The checks are the test files of Appendix E.

4. Results

4.1. What the closure contains

Table 3 gives the closure of each sort. Out of each member there are exactly as many symmetry operators as the point group has elements, and in every sort that number is the order of the holohedry: 48, 24, 16, 12, 8, 4 or 2.[z46] The twofolds, with the inversion, generate the group in every sort, so finding the twofolds suffices.[z46]

Table 3. *The Selling closure of each Delone sort, for a generic lattice of the pattern. Patterns are Selling parameters in the* S^6 *order* $b\cdot c, a\cdot c, a\cdot b, a\cdot d, b\cdot d, c\cdot d$*; equal letters denote equal values and* 0 *a zero. Members are counted up to swaps. "Visible" is the number of symmetries that carry the reduced superbase to itself; the rest carry it to another member. "Shapes" is the number of distinct cell shapes among the members, equal in every case to the number of orbits of members under the symmetries. "Visible" depends on which member one reduces to; values are for the pattern as written.*

sort	Bravais	pattern	zeros	members	|group|	visible	shapes
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr0rr0	2	3	48	16	1
C5	cP	000rrr	3	16	48	12	2
H4	hP	00rrrs	2	6	24	4	1
T1	tI	rrsrrs	0	1	16	16	1
T2	tI	rr0rrs	1	2	16	8	1
T5	tP	000rrs	3	16	16	4	3
R1	hR	rrrsss	0	1	12	12	1
R3	hR	rr0sr0	2	3	12	4	1
O1A	oF	rrsrrt	0	1	8	8	1
O1B	oI	rstrst	0	1	8	8	1
O2	oI	rs0srt	1	2	8	4	1
O3	oI	rs0rs0	2	3	8	8	2
O4	oS	00rsst	2	6	8	4	2
O5	oP	000rst	3	16	8	2	4
M1A	mC	rrsttu	0	1	4	4	1
M1B	mC	rstrsu	0	1	4	4	1
M2A	mC	rs0stu	1	2	4	2	1
M2B	mC	rs0rst	1	2	4	4	2
M3	mC	rs0ts0	2	3	4	4	2
M4	mP	00rstu	2	6	4	2	3
A1	aP	rstuvw	0	1	2	2	1
A2	aP	rs0tuv	1	2	2	2	2
A3	aP	rs0tu0	2	3	2	2	3

Zero pattern and sort fix different parts. Which members exist and which operators join them is fixed by the zero pattern; which of the operators are symmetries is fixed by the sort, and the abstract group by the Bravais type. Primitive monoclinic, C-centred orthorhombic and primitive hexagonal lattices all have two zeros sharing a vector and six members and differ only in how many operators are symmetries (4, 8, 24). Within one sort the shape of the cell plays no part: the members, the operators and the symmetries are the same, and the lattice supplies only the numbers attached to them.

Symmetry visible in one cell. In the sorts with walls the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive

tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal, 4 of 12 for rhombohedral R3.[z46] For the twofolds alone: none of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic; of the five, nine and seven of primitive tetragonal, cubic and hexagonal, one, three and one do.[z46] The rest are short flip words to other members.

Cell shapes. Two members are related by a symmetry exactly when they have the same shape: a symmetry preserves the metric tensor and hence the conorm multiset, and conversely the change of basis that carries one member's conorm vector onto another's preserves the metric tensor by equation (1). So the number of shapes equals the number of symmetry orbits (last column of Table 3), and no symmetry relates cells of different shape.[z46] Two measurements of the monoclinic lattice that reduce to M_0 and to M_1 report different cell parameters and are related by no symmetry, only by the flip at $a \cdot b$. Ten of the 24 sorts have more than one shape; primitive orthorhombic has four, the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal. Every extra shape looks like a conventional cell of some class, and the three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the catalogue is Table 1 of Appendix C. The same algebra gives the conditions under which a lattice of one sort becomes a lattice of another. Every such condition of codimension one is a conorm vanishing or two conorms becoming equal; the classical values $c/a = \sqrt{2}$, $b = \sqrt{3}\,a$ and $a = c$ in a monoclinic cell are among them, and the monoclinic conditions come in threes, one per cell choice (Table 2, Appendix C).[z50]

4.2. Near-congruence within a band

The measured problem. For the measured worked cell of Section 3.4, with $\alpha = 90.25°$, the conorms p_{ab} and p_{bc} are small numbers of either sign. The flips that gave M_1 and

M_2 now give superbases with a slightly negative conorm, Selling reduction rejects them, and the exact closure has one member instead of six; the twofold and the mirror, which were operators to M_4, now lead to a cell that is not in the set.

What the widened closure is. Equation (2) names no starting cell, order of operations or frame, so $\mathcal{S}_\tau(\mathbf{G})$ is the same whoever computes it. By Proposition A1 (Appendix A) a superbase belongs to it exactly when it is the exactly reduced superbase of some lattice within τ of $\mathbf{G}$ in that superbase's own conorms: the widened closure is the union of the exact closures of all lattices consistent with the measurement. The cell at which reduction with a $-\tau$ boundary stops is not well defined: from 400 random starting bases, reduction of the measured worked cell at $\tau = 43.2\,\text{Å}^2$ stopped at six different cells and of a noisy orthorhombic cell at sixteen, every one a member of $\mathcal{S}_\tau$, but on a noisy cubic cell at $\tau = 10\,\text{Å}^2$ the 400 restarts found 15 of the 16 members.[z45] The set has to be computed, not sampled.

Near-congruence is decided by the widened closure. If two cells are τ-congruent through a superbase F reduced for the new cell, then F, carried to the reference, has all conorms at least $-\tau$ and so lies in the reference's widened closure. Near-congruence is therefore decided without a search over operators: compute $\mathcal{S}_\tau(\mathbf{G}_R)$ once, reduce the new cell once, and pair its reduced superbase with each stored member. The stored set contains the connecting cell whenever one exists.

The computation is complete. If $\lambda_1 > 32\tau$, the four steps of Section 3.2 return exactly $\mathcal{S}_\tau(\mathbf{G})$ (Theorem A6 and Corollary A6a, Appendix A): every superbase within the band is an exact member of the idealised lattice, so enumerating the idealised closure and filtering cannot miss it. On every test lattice and band, including walls placed between τ and 4τ, the procedure returned the brute-force set.[z44] The condition is sufficient, not sharp: for a shortest cell edge of 30 Å it allows bands up to $28.1\,\text{Å}^2$, i.e. with $\tau = 3\eta$ a metric error of about $9\,\text{Å}^2$; the worked cell's error of $14.4\,\text{Å}^2$ is beyond

that, so at $\tau = 20\,\text{Å}^2$ the procedure is an instance of the theorem and returns six members, while its exactness at 43.2 and $60\,\text{Å}^2$ is an observation.[z44] Idealising all small conorms together is required by the proof and loses nothing, since the closure of any partial idealisation lies inside that of the full one (Lemma A8); the threshold 4τ is what the proof requires, and on dense two-wall grids 2τ was already exact while τ was not.[z46]

Scale and anisotropy. The condition $\lambda_1 > 32\tau$ is a ratio of two squared lengths, so it does not depend on the size of the cell: for an angular error σ the band on a conorm between vectors of lengths $|u_i|$, $|u_j|$ is about $\sigma|u_i||u_j|$, and for a cell with edges all near a the condition is $\sigma \lesssim 1/32$ rad, about 1.8°, for a 5 Å cell as for a 300 Å one. What it responds to is anisotropy: for a needle or a plate one uniform band must cover the conorms between the long vectors, and the condition fails at a few thousandths of a degree. With a band per slot, $\tau_{ij} = \sigma|u_i||u_j|$ (the covariance route of Appendix B), and the same four steps, needles are recovered — exact in all 44 cases in which the idealised lattice existed — while a plate is a genuine floor: beyond about 0.05° for $100 \times 100 \times 5$ Å the widened closure has more than sixteen members and is not the closure of any one lattice.[z49] Both failures are seen at step 2, before any member is proposed, and the brute-force window of Section 3.4 is the fallback; the per-slot statement is checked, not proved, and the numbers are in Appendix B.

Walks from the measured cell miss members. The natural alternative starts from the measured reduced cell and flips across any conorm within some band of zero. Table 4 compares four versions with the procedure above on 80 cells with walls at awkward positions (five zero patterns, including a plate-shaped orthorhombic cell), constructed as worst cases; all four miss members. The cause is a member reachable only through a non-member: with walls at 1.2τ and 0.5τ, the flip across the first gives a conorm of -1.2τ, outside the band, and only a second flip, across the other wall, reaches a

member of depth 0.7τ. On the idealised lattice both walls are zero and both flips cost nothing. Idealising at τ rather than 4τ fails on the same configuration.[z44] The most negative conorm of such a member is the difference of two small conorms, and we call it a *difference-wall member*; in the configurations checked they appear when one small conorm $z \leq \tau$ and another lies between τ and $z+\tau$; they carry no symmetry, and they are ordinary alternative reduced cells to which a second measurement of the lattice can reduce.[z46]

Table 4. *Four walks on the measured metric, and the procedure of Section 3.2, against brute force on 80 cells with walls at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns;* $\tau = 0.1$ *and* 0.5 *of* $\lambda_1/32$*).*

procedure	cells with a wrong set
flip wherever $\lvert p\rvert \leq \tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \leq \tau$, filter at the end	13
flip wherever $\lvert p\rvert \leq 2\tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \leq 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

The idealised lattice is not a symmetry assignment. Setting the small conorms to zero resembles the most generous choice a Le Page search makes, but Le Page returns a point group, asserted, whereas the idealisation returns the closure of a lattice Λ^* whose symmetries are candidates with prices, and the filter removes the members the measurement does not support. Idealisation imposes zeros but no equalities, so a measured primitive monoclinic cell whose only small conorms are its two walls idealises to a primitive monoclinic lattice; only when the small conorms are the walls of a higher class does Λ^* lie in that class, and then its extra symmetries are the pseudo-symmetries, each priced.

A worked filter. Table 5 shows the procedure on the measured worked cell. The reduced superbase has conorms $(49.5, 1588.9, 2444.7, 4.2, 847.5, 4.5)$ Å^2. At $\tau = 43.2$ Å^2 the three conorms below $4\tau = 172.8$ are set to zero and equation (1) gives an orthorhombic idealised lattice whose reduced superbase has edges 29.1, 39.9, 57.4 Å with $\beta = 120.5°$ — the reduced cell of an orthorhombic lattice is one of its monoclinic-

looking shapes; the orthogonal cell is $29.1 \times 39.9 \times 49.4\,\text{Å}$ on $(a, b, a + c)$. Its closure has sixteen members; each row gives one as a word in the flips of Table 1 and its six conorms under the *measured* metric. A member is kept when its smallest conorm is at least $-\tau$ and nothing is inherited: for $4.5 \le \tau < 8.7\,\text{Å}^2$ the member *ab.bc.db* is kept while its parent *ab.bc* is rejected, and for $45.0 \le \tau < 53.7$ the member *da.ab.bc* while its parent *da.ab* is rejected.[z45] Deeper members carry integer combinations of the small conorms — $-(49.5 + 4.2)$, $-(49.5 - 4.5)$ — and those combinations decide when a member enters. One enumeration serves every band up to $211\,\text{Å}^2$, where 4τ reaches the next conorm: six members for $8.7 \le \tau < 45.0$, among them the band $\tau = 20$ within the theorem, the six of the unperturbed cell at the same integer coordinates, and all sixteen from 62.5.[z45] The member entering at 45.0 carries the near-orthorhombic twofolds, so a member's depth is the band at which the pseudo-symmetry it carries becomes visible, and its symmetries' prices say how far the measurement is from it.

Table 5. *The filter on the measured worked cell. Each row is a member of the idealised lattice's closure, named by a word in the flips of Table 1 from the reduced cell e; the six numbers are its conorms under the measured metric tensor, slots da, db, dc, ab, ac, bc, in Å²; "enters at" is its depth. At τ = 43.2 the six members above the line are kept.*

word	*da*	*db*	*dc*	*ab*	*ac*	*bc*	enters at
e	49.5	1588.9	2444.7	4.2	847.5	4.5	0
ab	851.7	1593.1	2440.6	−4.2	53.7	8.7	4.2
bc	45.0	8.7	2449.3	1593.4	852.0	−4.5	4.5
ab.bc.db	2453.8	−4.5	847.5	1606.3	57.9	−4.2	4.5
ab.bc	842.9	4.5	2449.3	1601.8	62.5	−8.7	8.7
bc.db	2458.0	−8.7	53.7	1602.1	843.3	4.2	8.7
da	−49.5	2494.3	1638.4	53.7	897.0	−45.0	49.5
da.ab.bc	942.0	−45.0	1593.4	2556.7	12.9	−8.7	45.0
da.ab.ac	−49.5	2543.8	1588.9	954.9	−4.2	12.9	49.5
da.ab	950.7	2548.0	1584.7	−53.7	4.2	8.7	53.7
ab.ac	49.5	1539.3	2494.3	905.4	−53.7	62.5	53.7
bc.db.dc	45.0	2511.7	−53.7	1548.4	897.0	57.9	53.7
ab.ac.bc.ac	1610.5	959.1	2565.5	−4.2	−8.7	−53.7	53.7
da.ab.bc.ac	2569.6	−57.9	1606.3	954.9	−12.9	4.2	57.9
ab.ac.bc	−12.9	967.8	2556.7	1601.8	8.7	−62.5	62.5
ab.bc.ac	1664.2	−57.9	2511.7	905.4	−62.5	53.7	62.5

4.3. Reindexing from one reduced cell

The operators relating two settings of one lattice are one operator composed with the lattice symmetries, and each is an operator between a member of one closure and a member of the other. For the $\beta = 92^\circ$ cell and its perturbation at $\tau = 43.2\,\text{Å}^2$, pairing the six members of one with the six of the other, each pair in its best relative order, gives twelve pairs at a price of $14.4\,\text{Å}^2$ and nothing else below $1378\,\text{Å}^2$; the twelve pairs are four distinct operators, one operator composed with each element of $2/m$, at a price equal to the perturbation. At $\tau = 60\,\text{Å}^2$ the members carrying the near-orthorhombic symmetry are present and add ten pairs at $150.7\,\text{Å}^2$, so the band decides which pseudo-symmetries are offered.[z39]

A new cell needs no closure of its own. For fifteen references near walls of five zero patterns, each with twelve new cells carrying a random metric error of declared size and a random reindexing, the new cell's single reduced superbase was always a member of the reference's widened closure at the correct integer coordinates, and the true operator was always among the candidates: 180 of 180, with the band set from the declared error by the coverage bound of Appendix B.[z47]

Completeness of the stored set matters. A new cell constructed so that its only reduced superbase is a difference-wall member of an orthorhombic corner was matched against two stored sets of the reference. The set built as in Section 3.2 offers candidates at 7.83, 8.01 and $8.84\,\text{Å}^2$, each with its negative; the set built by idealising only at τ offers the first two and has lost the third, which is $\pm\mathbf{I}$, the true operator.[z46] The lowest-priced candidate is found either way, through a pseudo-symmetry, but it is not the true operator, which is the one lost; price orders the coset, it does not identify its true member.

The band that makes the true operator one of the candidates can be stated in advance, under a hypothesis on the error expressed in the reference frame. Suppose

the new cell's metric tensor, carried to the reference frame by the true operator, differs from the reference's by at most η in Frobenius norm. Let V be the sum of the squared lengths of the new cell's three reduced vectors and $\lambda_{\min}$ the smallest eigenvalue of the reference metric tensor. Then the band $\tau = \sqrt{2}\,\eta\,V/(\lambda_{\min}-\eta)$ suffices, provided $\eta < \lambda_{\min}$ and $\lambda_1 > 32\tau$ (Proposition A10, Appendix A). Appendix B gives a bound that does not refer to the operator. The guarantee held in all 180 cases but is conservative, with median 13η against a band near 3η that was in fact enough.[z47]

5. Discussion

Relation to earlier work. The ingredients are classical. Selling (1874) gave the reduction and the identity of Appendix A; Voronoi (1908) the relevant vectors; Delaunay (1933) the 24 sorts and the observation, made practical by Patterson & Love (1957) and tabulated in *International Tables* (Burzlaff & Zimmermann, 2006), that the reduced superbase is not unique when a conorm vanishes. Conway & Sloane (1992) supplied the conorm language and the theorem that a superbase is obtuse exactly when its partial sums are the Voronoi vectors, with the five zero patterns as the five types of Voronoi polyhedron. Kurlin (2022) enumerated the obtuse superbases of each pattern. Andrews, Bernstein & Sauter (2019a,b; Andrews & Bernstein, 2023a) cast the same objects as S^6 with its boundary transformations. On the Niggli side the multiplicity of Buerger cells (Buerger, 1960; Gruber, 1973) and the instability of the reduced cell under error (Andrews *et al.*, 1980; Andrews & Bernstein, 1988) are the phenomenon treated here, and Gruber's (1997, 2006) 127 genera refine both the characters and the sorts. Symmetry from the metric tensor with a tolerance is Le Page (1982), Himes & Mighell (1987) and Sauter *et al.* (2004); the operators relating two settings form a coset of the symmetries (Flack, 1987). Closest in aim is Oishi-Tomiyasu's error-stable Bravais-lattice determination (2012, 2024), alongside the lattice-character

quality indices of Kabsch (1993). She gives a bound on the error under which a finite list of basis transforms is provably enough, which is the role $\lambda_1 > 32\tau$ plays here, and she names the failure that the difference-wall members of Section 4.2 resolve: a noisy cell that is nearly reduced while the true cell is not reduced in the same basis. On the seven measured cells she distributes, the prices of the closure's operators, taken in increasing order, reproduce her ranking of Bravais types (Appendix C). What is added here is the accounting of Table 3, which separates the symmetries visible in a reduced cell from those that are operators to other cells; the derivation of the sorts as closed sets of equalities; the widened closure with its uniqueness and union characterisation; the idealisation at 4τ with its completeness theorem and the demonstration that walks from the measured cell are incomplete; and the reindexing guarantee from one reduced cell.

Where reindexing has been hard. Within one sort, two correct descriptions of a lattice can differ in shape. Each is a reduced cell that Selling reduction, or Niggli reduction with a tolerance (Křivý & Gruber, 1976; Grosse-Kunstleve *et al.*, 2004), can return when a near-zero conorm falls on the other side. No symmetry relates them, so a comparison that applies only the symmetries of the cell it holds cannot reconcile them. The monoclinic c versus $a + c$ choice is the oldest instance; the monoclinic-looking cells of orthogonal lattices are the same phenomenon. Across sorts it is the walls: a cell measured within τ of a wall idealises into a sort with more members, each entering at a depth and carrying a priced pseudo-symmetry. The widened closure holds both kinds of alternative with a depth for each, so under the condition of Theorem A6 a reindexing is a lookup rather than a search whose completeness depends on the cell it started from. The instability of the Niggli cell (Andrews *et al.*, 1980; Andrews & Bernstein, 1988) is the same phenomenon from the Niggli side: in the benchmark of Appendix C the measured cell's Niggli cell is a different cell from the reference's in

45 of 780 cases, nearly all at the body-centred orthorhombic sort O3 — a sort whose Bravais type also contains sorts with fewer zeros — and its neighbours, where the two Niggli cells are two shapes of one closure related by a flip and by no symmetry.

What is proved and what is checked. Proved in Appendix A, from Selling's identity, the Conway–Sloane characterisation of obtuse superbases, Kurlin's enumeration and two integer constants obtained by exhaustive enumeration, are the uniqueness and union characterisation of the widened closure, the completeness of idealise–enumerate–filter when $\lambda_1 > 32\tau$, the generation of the closure by flips and swaps, the nesting of partial idealisations, and the reindexing guarantee. The sorts as closed sets of equalities (Proposition A9) is a definition whose agreement with Delone's 24 is checked by enumeration, and the counts of Table 3 are exhaustive on generic lattices with random values. Kurlin's enumeration is cited from a preprint; his counts are reproduced here by brute force for every sort. Checked in every case tested but not proved are that idealising at 2τ suffices, that a band per slot recovers anisotropic cells whenever the idealised lattice exists, and that the true operator is found at the band set from the error rather than only at the conservative guarantee.

The frame. Statements about the widened closure are in terms of τ, a band on conorms, and need no frame. Converting an error on the metric tensor into τ does, because the Frobenius norm of an error depends on the basis; Appendix B gives the exact relation between a metric error and each conorm shift, the worst-case factors by zero pattern (up to $2\sqrt{3}$ in some cubic frames, so that $\tau = 3\eta$ is not a worst-case guarantee everywhere), a frame-free bound and the covariance route.

Benchmark. The places where the closure predicts trouble — the shapes, lattices a small distance from a specialisation condition, cells whose only reduced superbase is a difference-wall member, anisotropic cells, the sorts O3, M2B and M3 — are collected as a benchmark of 780 measured cells with their true operators (Appendix C). On it

a symmetry-only reindexing proposes the true operator in 54% of the cases, the exact closure in 96%, and a walk at τ in 97%, with an incomplete set on every difference-wall case. The procedure of Section 3.2 declines 8 anisotropic cases at step 2 and proposes the true operator in all of the other 772.

6. Conclusions

The Selling closure of a lattice contains, in one finite and complete set, every symmetry and every operator between alternative reduced cells; a single reduced cell shows only the symmetries that relabel it, and no symmetry relates cells of different shape. For a measured cell, near-congruence within a band is decided by the widened closure, which is unique and is computed completely, when $\lambda_1 > 32\tau$, by idealising at 4τ, enumerating and filtering; the condition is a ratio of lengths, independent of the scale of the cell, and its failure for strongly anisotropic cells is seen before any operator is proposed. Reindexing a new cell onto a reference then reads one operator from that set, from a single reduced cell of the new cell, and when the error is below the smallest eigenvalue of the reference metric and the band it implies satisfies the same condition, the true operator is guaranteed to be among those read.

Appendix A
Proofs

Terms. Write a lattice vector in any basis, $x = n_1 v_1 + n_2 v_2 + n_3 v_3$ with integer n_i, and record whether each coordinate is even or odd. This sorts the lattice into eight *parity classes*; two vectors are in the same class exactly when their difference is twice a lattice vector, so the classes do not depend on the basis (the classes of $\Lambda/2\Lambda$). One class is the doubled vectors; the other seven are used here. A *partial sum* of a

superbase is the sum of some of its four vectors; since the four add to zero, there are seven partial sums up to sign, one in each of the seven classes, and for the monoclinic superbase $(-a-b-c, a, b, c)$ they are a, b, c, $a+b$, $a+c$, $b+c$ and $a+b+c$. The fact used is that a superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, the Voronoi vectors of the lattice (Voronoi, 1908; Conway & Sloane, 1992, whose vonorms are these minima; see also Kurlin, 2022, Lemmas 2.4, 2.7 and A.1). The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum $v_c(F)$ in that class minus the squared length μ_c of the shortest vector of the class; an obtuse superbase has zero excess in every class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$ marking which vectors it adds; a vector of the same parity class has coordinates $s + 2k$ with k integer. Every lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{3}$$

whatever the signs of the conorms, because the four vectors sum to zero. For a superbase F and two metrics, $\delta_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition A1 (uniqueness and union).

$$\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{C}(\mathbf{G}') \text{ for some } \mathbf{G}' \text{ with } \delta_F(\mathbf{G}, \mathbf{G}') \le \tau\}.$$

Proof. ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\delta_F(\mathbf{G}, \mathbf{G}') \le \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) If $F \in \mathcal{S}_\tau(\mathbf{G})$, the metric whose conorms in the frame of F are $\max(p_{ij}, 0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite; F is obtuse for it, at δ_F equal to the depth of F. Each negative conorm must move at least to zero, so no nearer metric makes F obtuse. □

Lemma A2 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial sum and w' a vector of its parity class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + d)$, $\Delta k = k_i - k_j$, $d = s_i - s_j$; each γ_{ij} is a non-negative

integer, zero exactly when $\Delta k \in \{0, -d\}$. *Proof.* Equation (3), $(d + 2\Delta k)^2 - d^2 = 4\Delta k(\Delta k + d)$, and $t(t + d) \geq 0$ for integers t and $d \in \{-1, 0, 1\}$.

Lemma A3 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\leq \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \geq 4m^*$ in some class. *Proof.* In the frame of F_0, with w the partial sum of F_0 in class c — a shortest vector of its class for $\mathbf{G}$ and for Λ^* — Lemma A2 gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, a sum of non-negative terms, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, every partial sum of F is a shortest vector of its class for Λ^* and F is obtuse for it; otherwise some $\gamma_{ij} \geq 1$ multiplies a $p^0_{ij} \geq m^*$.

Lemma A4 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \leq 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \leq 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. With w' a shortest vector of the class, Lemma A2 in the frame of F gives $0 \leq e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \leq \tau \sum_N \gamma$. If a large γ were positive, Lemma A5(ii) would give $8\tau \sum_{\text{large}} \gamma < \tau \sum_N \gamma \leq 8\tau \sum_{\text{large}} \gamma$; so the large γ vanish, and Lemma A5(i) gives $e_c \leq 4\tau \cdot 4$.

Lemma A5 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \leq 4$. (ii) If $\sum_L \gamma_{ij} > 0$, then $\sum_N \gamma_{ij} \leq 8 \sum_L \gamma_{ij}$. Both bounds are attained. *Proof.* In (i) $|\Delta k| \leq 1$ on L, so with $k_0 = 0$ all $|k_i| \leq 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \leq 15$; beyond, $Q = \sum_L \Delta k^2 > 75$, and $\sum_N \gamma \leq \rho Q + \sqrt{|N|\rho Q}$, $\sum_L \gamma \geq Q - \sqrt{|L|Q}$, with $\rho \leq 3 + 2\sqrt{2}$ the largest generalised eigenvalue of $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$, keep it below 8 for every pattern. Equality in (ii): $L = \{01, 02, 13\}$,

$N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$.[z44]

Theorem A6. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{C}(\Lambda^*) : p_{ij}(F; \mathbf{G}) \geq -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma A4 gives $e_c(F; \mathbf{G}) \leq 16\tau$ in every class. The excess of F for Λ^* is the same sum of non-negative terms with the zeroed slots dropped, so $e_c(F; \Lambda^*) \leq e_c(F; \mathbf{G}) \leq 16\tau < 4m^*$, since the smallest surviving conorm m^* exceeds 4τ; Lemma A3 then makes F obtuse for Λ^*. □

Corollary A6a (operator form). Under the hypotheses of Theorem A6 let $\mathbf{W}_k$, $k = 1, \ldots, m$, be the integer changes of basis from the basis of F_0 to the members of $\mathcal{C}(\Lambda^*)$. Then

$$\mathcal{S}_\tau(\mathbf{G}) = \{F_0\mathbf{W}_k : \min_{ij} p_{ij}(F_0\mathbf{W}_k; \mathbf{G}) \geq -\tau\},$$

where $F_0\mathbf{W}_k$ is the superbase whose basis is that of F_0 transformed by $\mathbf{W}_k$. *Proof.* The members of $\mathcal{C}(\Lambda^*)$ are the superbases $F_0\mathbf{W}_k$; the $\mathbf{W}_k$ are integer matrices independent of the metric, so the same superbases read with $\mathbf{G}$ are the right-hand side of Theorem A6. □

Lemma A7 (generation). Every operator between members of the closure, and every symmetry, is a sequence of flips at zero conorms followed by a swap. *Proof.* The members are reached from F_0 by flips at zero conorms (Kurlin, 2022, Lemmas 4.1–4.5), so flips and swaps connect all members. A symmetry maps F_0 to an obtuse superbase, which is a member; as a matrix it is the flip path from F_0 to that member followed by a swap.

Lemma A8 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The members for Z' are reached from F_0 by flips at conorms that are zero for Z' (Lemma A7), and a flip

at a zero conorm exchanges two conorms and leaves the rest (Section 2.2), so every member carries the conorm values of F_0 in some order. Those conorms are zero for Z as well, so the same flips are flips at zeros for Z, and every member for Z' is a member for Z.[z46]

Proposition A9 (sorts as closed sets of equalities). Fix a zero pattern Z, let O_Z be the operators of its closure and, for $\mathbf{W} \in O_Z$, let $L_{\mathbf{W}}$ be its linear action on conorm vectors (Appendix D), which on the face $\{p \geq 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace V of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : L_{\mathbf{W}} = \mathrm{id} \text{ on } V\}$ and $\mathrm{Fix}(H) = \{p : L_{\mathbf{W}} p = p \ \forall\, \mathbf{W} \in H\}$. A lattice with conorm vector p on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in O_Z by Lemma A7, and the lattices with the symmetries of a generic point of V fill $\mathrm{Fix}(\mathrm{Sym}(V))$. We take as definition that a sort is a maximal set of lattices with the same zero pattern and the same symmetry operators; then the sorts of zero pattern Z are the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the operators of O_Z that map the face onto itself. *Proof.* Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$, $\mathbf{W} \in O_Z$; since $L_{\mathbf{W}}$ permutes the slots on the face, each $\mathrm{Fix}(\mathbf{W})$ and each intersection is a set of equalities among conorms. A closed subspace with an interior point is a maximal set of lattices with the same symmetries, and two such subspaces exchanged by an operator preserving the face describe the same lattices from another member. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three, with symmetry orders $2, 4, 4, 8, 8, 12, 16, 48$; $2, 4, 4, 8, 16$; $4, 8, 24$; $2, 4, 8, 12, 48$; $8, 16, 48$; these are Delone's 24 sorts with their holohedries, as listed in Table 3, which is the check that the definition is his.[z53]

Proposition A10 (reindexing guarantee). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \leq \eta < \lambda_{\min}(\mathbf{G}_R)$, let F_N be its reduced superbase

with basis vectors of summed squared length V, and let $\tau \geq \sqrt{2}\,\eta\,V/(\lambda_{\min}(\mathbf{G}_R) - \eta)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, the procedure of Section 3.2 produces it, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* Let u_i be the coordinates of F_N's basis vectors in the reference frame, the rows of $U = \mathbf{P}^\mathsf{T}$ for the true operator $\mathbf{P}$, and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$. Then $u_i^\mathsf{T}\mathbf{G}'_N u_i \geq (\lambda_{\min}(\mathbf{G}_R) - \eta)|u_i|^2$, so $\sum |u_i|^2 \leq V/(\lambda_{\min} - \eta)$. In F_N's own frame the error is $U\Delta U^\mathsf{T}$, of Frobenius norm at most $\sum |u_i|^2\,\eta$, and in its own frame every conorm shift is at most $\sqrt{2}$ times that norm (Appendix B). The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem A6 applies. The proposed operator for the pair is the change of basis between the two copies of F_N, which is the true one. □

Appendix B
From a metric error to a band

A conorm is linear in the metric, so a metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by exactly $-u_i^\mathsf{T}\Delta\mathbf{G}\,u_j$; by Cauchy–Schwarz in the Frobenius product the shift is at most $c_{ij}\eta$, with $c_{ij} = \|\operatorname{sym}(u_i u_j^\mathsf{T})\|_F = [\frac{1}{2}(|u_i|^2|u_j|^2 + (u_i \cdot u_j)^2)]^{1/2}$, and an error along $\operatorname{sym}(u_i u_j^\mathsf{T})$ attains it. Every exact member of a stored closure survives in the widened closure of the measurement, for every error of norm η, exactly when τ is at least the coverage bound $\tau_{\text{cov}} = \max(c_{ij}\eta - p_{ij})$, read off the stored closure.[z36] For the worked cells $\eta = 14.4\,\text{Å}^2$ and $\tau = 3\eta = 43.2\,\text{Å}^2$. Member vectors are not confined to $\{-1, 0, 1\}$; on two and three walls they carry entries up to 3. On walls the largest constant is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and with three zeros $2\sqrt{3}$ in six of the sixteen labelled cuboid frames — those whose zeros form a path with d at an interior vertex — and at most $\sqrt{5}$ otherwise; on positive slots it is at most 7.2. In a $2\sqrt{3}$ frame an adversarial error of norm η drives a member

to $-3.46\,\eta$, so $\tau = 3\eta$ is not a worst-case guarantee in every frame.[z36] In a member's own frame the member is the standard superbase, whose largest constant is $\sqrt{2}$, so $\tau = \sqrt{2}\,\eta^*$ with η^* the largest error norm over the members' own frames covers every member, and $\eta^* \le \eta \max_k \|U_k\|_F^2$ over the stored members' coordinate matrices U_k, which is computable from η alone; it is uniform but loose, 2.8 times the reduced-frame η in the median case and up to 17 times.[z36] With a covariance Σ for the metric, the variance of each member conorm is the quadratic form of Σ on $\mathrm{sym}(u_i u_j^{\mathsf{T}})$, exact in any frame, and the band can be set slot by slot. With $\tau_{ij} = \sigma|u_i||u_j|$ and the four steps of Section 3.2, three needles ($5 \times 5 \times 100$ Å monoclinic and orthorhombic, $40 \times 40 \times 300$ Å hexagonal; four random errors at each of 0.02°, 0.05°, 0.2°, 0.5°) returned the brute-force set in all 44 cases in which the idealised lattice existed; a $100 \times 100 \times 5$ Å plate was exact at 0.02° and 0.05° but at 0.2° and 0.5° its widened closure has 24 and 36 members, more than any closure of a single lattice, and the condition $\lambda_1 > 32 \max \tau_{ij}$ over the idealised slots fails there, as it does for $\sigma > (a_{\mathrm{short}}/a_{\mathrm{long}})^2/32 = 0.0045^\circ$ by the uniform estimate.[z49]

Appendix C
Catalogues

Cell shapes. Table 1 lists the shapes of the ten sorts with more than one, each as the cell with the most right angles and no acute angle. The three shapes of a primitive monoclinic lattice are the three cells on $\{a, c, a + c\}$ with b unique, the cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the extra shapes of primitive orthorhombic, tetragonal and cubic lattices are monoclinic-looking cells with two right angles; C-centred orthorhombic has two primitive shapes; the boundary sorts of body-centred orthorhombic, C-centred monoclinic and triclinic have two or three cells with

a single right angle each, for triclinic the known instability of the reduced cell when an angle is near 90° (Andrews *et al.*, 1980).[z49]

Table 1. *The cell shapes of the sorts with more than one, for the cells named; each shape is given as the cell with the most right angles and all angles* $\geq 90°$*, shortest edges first when several qualify and b unique for the monoclinic-looking ones, with the number n of members (up to swaps) that have it. Centred cells are given by their primitive cells; the boundary sorts are specified by Selling parameters in the* S^6 *order of Table 3, in* $Å^2$*.*[z49]

sort	lattice	n	shape
cP	30 Å	4	30, 30, 30; 90, 90, 90
		12	30, 30, 42.4; 90, 135, 90
tP	30, 30, 50 Å	4	30, 30, 50; 90, 90, 90
		4	30, 30, 58.3; 90, 121.0, 90
		8	30, 50, 42.4; 90, 135, 90
oP	30, 40, 55 Å	4	30, 40, 55; 90, 90, 90
		4	30, 40, 62.6; 90, 118.6, 90
		4	30, 55, 50; 90, 126.9, 90
		4	40, 55, 50; 90, 143.1, 90
oS	C 30, 40, 50 Å	2	25, 50, 25; 90, 106.3, 90
		4	25, 50, 30; 90, 126.9, 90
mP	30, 40, 50 Å, 100°	2	30, 40, 50; 90, 100, 90
		2	30, 40, 53.7; 90, 113.4, 90
		2	50, 40, 53.7; 90, 146.6, 90
oI (O3)	900, 1600, 0, 900, 1600, 0	1	50, 50, 50; 111.1, 129.8, 90
		2	42.4, 50, 50; 90, 115.1, 115.1
mC (M2B)	900, 1600, 0, 900, 1600, 2500	1	50, 50, 70.7; 104.7, 116.9, 90
		1	50, 50, 65.6; 105.9, 105.9, 90
mC (M3)	900, 1600, 0, 2500, 1600, 0	1	50, 50, 64.0; 120, 90, 111.1
		2	50, 56.6, 58.3; 90, 108.0, 124.4
aP (A2)	900, 1600, 0, 2500, 400, 1200	1	36.1, 60.8, 64.0; 114.3, 90, 114.2
		1	36.1, 56.6, 64.0; 116.2, 90, 101.3
aP (A3)	900, 1600, 0, 2500, 400, 0	1	36.1, 50, 53.9; 90, 101.9, 119.9
		1	44.7, 50, 53.9; 90, 99.6, 135.7
		1	36.1, 44.7, 58.3; 90, 115.3, 104.4

Specialisations. The conorms of any member, in any labelling, are integer combinations of the reduced cell's conorms, hence linear in the free parameters of the sort (the letters of Table 3); imposing another sort's zeros and equalities on one member is a homogeneous linear system in those parameters, and a solution with non-negative parameters, a non-singular lattice and a resulting lattice of the target sort and no higher is a specialisation. Table 2 lists the 37 conditions of codimension one, on 36 pairs of sorts.[z50] Each is a parameter vanishing or two parameters becoming equal. The classical values appear: body-centred tetragonal is face-centred cubic at $c/a = \sqrt{2}$ and body-centred cubic at $c/a = 1$; C-centred orthorhombic is hexagonal at $b = \sqrt{3}\,a$

and tetragonal at $b = a$; rhombohedral is cubic at $\alpha = 60°$, $90°$ and $109.47°$. The monoclinic conditions come in threes, one per shape: with unique axis c and plane lattice a, b, the lattice is C-centred orthorhombic when $|a| = |b|$, $|a| = |a + b|$ or $|b| = |a + b|$, and primitive orthorhombic when $a \perp b$, $a \perp (a+b)$ or $b \perp (a+b)$ — the textbook condition $a = c$ read in each of the three cell choices. The table also shows which sorts are one condition from a much higher symmetry: C-centred monoclinic from tetragonal, body-centred tetragonal and rhombohedral; the body-centred orthorhombic boundary sort from face-centred cubic; triclinic on a wall from C-centred monoclinic.

Table 2. *All specialisations of codimension one between the 24 sorts. Conditions are on the letters of the sort's pattern in Table 3; "shapes" is the number of distinct cell shapes of the starting sort through which the conditions are reached, and the last column reads the condition in the cell of the conventional description where that is short.*[z50]

from	to	conditions	shapes	reading
aP A1	aP A2	any letter $= 0$	1	a right angle appears
aP A2	aP A3	$v = 0$	1	a second right angle
aP A2	mC M2A	$s = t$, $r = u$	1	
aP A2	mP M4	r, s, t or $u = 0$	1	
aP A3	mC M3	any two letters equal	3	one per shape
mP M4	oP O5	$r = 0$, $s = 0$, $t = 0$	3	$a \perp b$, $a \perp (a+b)$, $b \perp (a+b)$
mP M4	oS O4	$s = t$, $r = t$, $r = s$	3	$\lvert a\rvert = \lvert b\rvert$, $\lvert a\rvert = \lvert a+b\rvert$, $\lvert b\rvert = \lvert a+b\rvert$
mC M1A	oF O1A	$r = t$	1	
mC M1A	oI O2	$u = 0$	1	
mC M1A	oS O4	$r = 0$, $t = 0$	1	
mC M1B	mC M2B	$t = 0$, $u = 0$	1	boundary sort
mC M1B	mC M3	$r = 0$, $s = 0$	1	boundary sort
mC M1B	oF O1A	$r = s$	1	
mC M1B	oI O1B	$t = u$	1	
mC M2A	oI O2	$r = t$	1	
mC M2B	oI O3	$t = 0$	1	
mC M2B	tI T2	$r = s$	1	one step to tI
mC M2B	tP T5	$r = 0$, $s = 0$	1	one step to tP
mC M3	oI O3	$r = t$	1	
mC M3	hR R3	$r = s$, $s = t$	1	one step to hR
oP O5	tP T5	$r = s$, $r = t$, $s = t$	1	$a = b$, $a = c$, $b = c$
oS O4	tP T5	$r = 0$	1	$b = a$
oS O4	hP H4	$r = s$	1	$b = \sqrt{3}\,a$
oI O1B	oI O3	r, s or $t = 0$	1	boundary sort
oI O1B	tI T1	$r = s$, $r = t$, $s = t$	1	two edges equal
oI O2	tI T2	$r = s$	1	
oI O3	cF C3	$r = s$	1	one step to cF
oF O1A	tI T1	$s = t$	1	
oF O1A	tI T2	$s = 0$, $t = 0$	1	
tP T5	cP C5	$r = s$	1	$c = a$
tI T1	cI C1	$r = s$	1	$c/a = 1$
tI T1	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from below
tI T2	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from above
hR R1	cP C5	$r = 0$	1	$\alpha = 90^\circ$
hR R1	cI C1	$r = s$	1	$\alpha = 109.47^\circ$
hR R3	cF C3	$r = s$	1	$\alpha = 60^\circ$

Measured cells from powder indexing. Oishi-Tomiyasu (2024) distributes seven measured primitive cells with her program's output, a ranking of Bravais types by distance in the manner of the quality indices of Kabsch (1993): for each Bravais type, the nearest cell of that type and its distance. Table 3 treats each by Section 3.2 at $\tau = \lambda_1/32$ and prices every operator out of the reduced cell in the closure of the idealised lattice; the price at which the number of operators reaches the order of a holohedry is the

band at which that symmetry is offered.[z51] In all seven cases the band idealises the walls of the indexed type, and the type's holohedry is reached at a price between 0.9 and 1.8 times the program's distance, with the same ordering of types; the two measures differ in definition, so they agree in order and size, not in value. Her test set of 4738 cells is not distributed.

Table 3. *The seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024) by the closure: Bravais type and name as distributed; members of the idealised closure at $\tau = \lambda_1/32$; the price in $Å^2$ at which the number of operators out of the reduced cell reaches the order of the type's holohedry, and of the next holohedry up where one is reached; the program's minimal distance to a cell of that type.*[z51]

type	sample	members	price, type	price, next	distance
tI	HERMES_Sr327_250K	2	0.77	132.6	0.57
tP	1339prbasrcuo_bs1	16	0.10	26.0	0.056
cI	sample3	1	2.98	–	2.07
cF	CeO2_matsushita	3	0.11	–	0.084
hP	M402-pf-119673	6	22.5	196.1	20.5
mC	Acrinol-B-D8CuKa1	1	0.96	86.2	0.72
hR	Alumina676	3	1.69	9.18	1.93

Benchmark. The repository directory `manuscript/benchmark` holds 780 measured cells with exact references and true operators, in five families: 75 cells measured in a different shape of the same lattice; 660 lattices placed 0.5 to 8 times the error from a condition of Table 2; 20 cells whose only reduced superbase is a difference-wall member; 16 needles and plates with angular errors; 9 cells of the boundary sorts. Each case also carries its Niggli cell. An evaluator scores whether a method proposes the true operator from one reduction of the measured cell; the results quoted in Section 5 are in `results.txt`.

Appendix D
The operators alone

The closure can be described without any lattice. An operator $\mathbf{W}$, written as the new basis in the coordinates of the reduced cell, acts on the six conorms by a fixed

integer 6×6 matrix $L_{\mathbf{W}}$, since every conorm of $\mathbf{W}F_0$ is an integer combination of the conorms of F_0. The reduced cells are the points of the orthant $\{p \geq 0\}$ in conorm space; Selling (1874) showed that every lattice has a conorm vector in this orthant, and it is unique up to the 48 swaps except on the walls (Delaunay, 1933; Conway & Sloane, 1992), and the closure of a lattice is the set of points of its orbit that lie in the orthant: one in the interior, several on the walls. So $\mathbf{W}$ is an operator of the closure of every lattice with zero pattern Z exactly when $L_{\mathbf{W}}$ maps the face $\{p \geq 0,\ p_Z = 0\}$ into the orthant, which is a sign condition on the entries of $L_{\mathbf{W}}$ and nothing else. Enumerating integer matrices under that condition gives the operator set of each face without a lattice: 48×1, 2, 3, 6 and 16 matrices for no zero, one zero, two zeros on disjoint edges, two zeros on a path and three zeros, and their union, over all placements of the faces, is a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices with entries in $\{-3, \ldots, 3\}$ (whether the 127 coincides with Gruber's (1997) 127 genera of lattices we have not examined). $\mathcal{U}$ is closed under inversion, every element has a labelling whose basis vectors have entries in $\{-2, \ldots, 2\}$, and $\mathcal{U}$ contains every operator between two reduced cells of one lattice, whatever the lattice, since an operator out of any member is an operator out of a reduced cell of that member's zero pattern.[z52] The three-zero face enumerated is the triangle; its path placement gives the same operators relabelled. Up to relabelling at both ends $\mathcal{U}$ has nine kinds of element, of flip length 0 to 4 (Table 1).

Table 1. *The nine kinds of operator between Selling-reduced cells, up to relabelling at both ends; "matrices" counts the integer matrices of the kind in $\mathcal{U}$, "flips" its length as a flip word, and the last column the zero patterns in whose closures it occurs.*[z52]

kind	matrices	flips	occurs with
1	48	0	every pattern (the swaps)
2	288	1	every pattern with a zero
3	1152	2	two zeros on a path; three zeros
4	1152	3	three zeros
5	1152	3	three zeros
6	576	3	two zeros on a path; three zeros
7	576	4	three zeros
8	576	4	three zeros
9	576	4	three zeros

Why four. For a lattice with $a \perp b \perp c$ the sixteen members are four orthogonal

superbases, differing by the sign of one axis, and twelve with a face diagonal. A flip from an orthogonal cell always gives a cell with a face diagonal, so changing the sign of one axis takes three flips, out, across and in, and the diagonal cells attached only to the far orthogonal cell take one more: from (a, b, c) the members lie at flip distances $0, 1, 2, 3, 4$ in numbers $1, 3, 6, 3, 3$, and four is the diameter of the largest face.[z46]

Appendix E
Computational checks

Each mark [*znn*] refers to `manuscript/appendix_z/test_z`*nn*`_*.py` in the `agentsg` repository.

mark	what is checked
z36	shift identity; constants by zero pattern and frame; adversarial loss at 3η; frame-free bound
z39	measured worked pair: widened closure by definition; candidate operators at $\tau =$ 43.2 and 60
z44	constants 4 and 8; class gap and excess; Theorem A6 against brute force; grey zone; worked cell
z45	reduction endpoints and restarts; walks on the measured metric; the worked filter table
z46	the monoclinic example; closures of the 24 sorts; shapes = orbits; twofolds generate; coset completeness; difference-wall members; 2τ; nesting
z47	reindexing under a declared error: membership, true operator, guarantee
z48	the 24 Delone sorts reproduced by closed tie patterns on generic lattices
z49	bands per slot on needles and plates: exactness, the plate floor, detection of the failed hypothesis; the cell shapes of Table 1
z50	the 37 codimension-one specialisations between sorts and the shapes that carry them (Table 2)
z51	the seven BLDConograph sample cells: idealised closure, price ladder, comparison with the program's distances (Table 3)
z52	the operator sets of the faces from the matrices alone; the universal set $\mathcal{U}$, its inversion closure and window; the nine kinds (Table 1); universality over the 24 sorts
z53	the 24 sorts as the closed sets of equalities of Proposition A9, with their symmetry orders

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Synopsis

The set of all Selling-reduced cells of a lattice, with the integer matrices between them, holds the lattice symmetry and every alternative choice of reduced cell in one complete set. For a measured cell the set is recovered exactly by setting every small conorm to zero, enumerating and filtering, and reindexing reads one matrix from it.